Generative AI Applications for Managers

Estimated time: 30 minutes

Welcome to the Generative AI Applications for Managers.

Learning objectives

After completing this lab, you will be able to:

- Explore how generative AI can be used to get insights and recommendations for leaders and managers.
- Demonstrate examples of using generative AI for creating dashboards.

Introduction

Leaders and managers are increasingly leveraging generative AI to enhance decision-making processes and streamline operations within their organizations. By harnessing the power of advanced language models, leaders can utilize natural language understanding and generation to automate routine tasks, analyze vast amounts of data for actionable insights, and generate human-like text for various applications. Generative AI supports leaders in formulating creative solutions, improving communication, and fostering innovation by providing valuable assistance.

In this lab, you will work with ChatGPT and Google Spreadsheet to demonstrate the use of generative AI by managers.

> If you have access to Microsoft Excel, you can consider working in an Excel Workbook, instead of Google Spreadsheet.

Accessing Google Spreadsheets
To work with Google Spreadsheets, you need to have a Google account. If you do not have a Google account, then visit the following page and follow the steps to create an account.

```
https://support.google.com/accounts/answer/27441?hl=en
```

Log in to ChatGPT

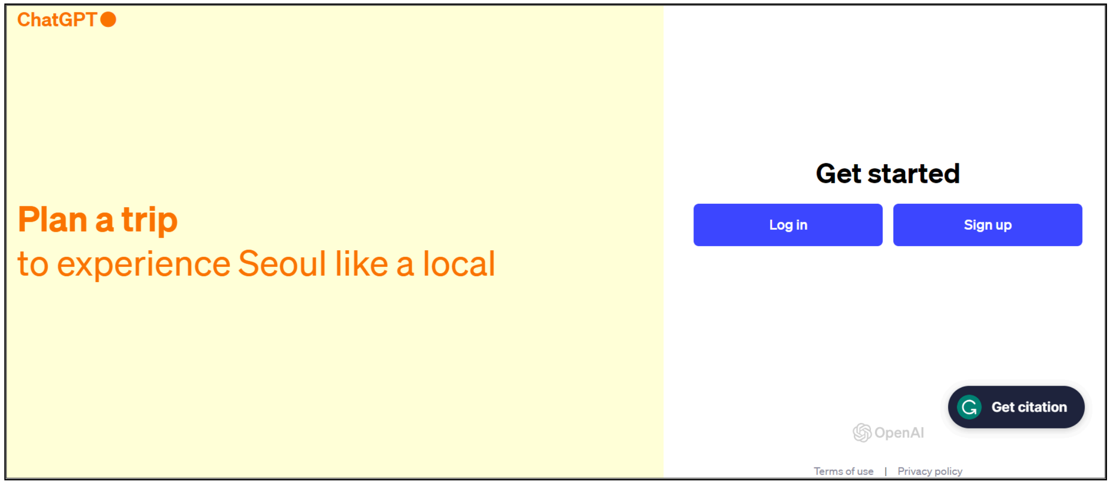
1. Launch [ChatGPT](https://chat.openai.com/) and sign in.

   > Note: If you cannot access ChatGPT by clicking on the link, copy the following URL and paste it into a browser.

   ```
   https://chat.openai.com/
   ```

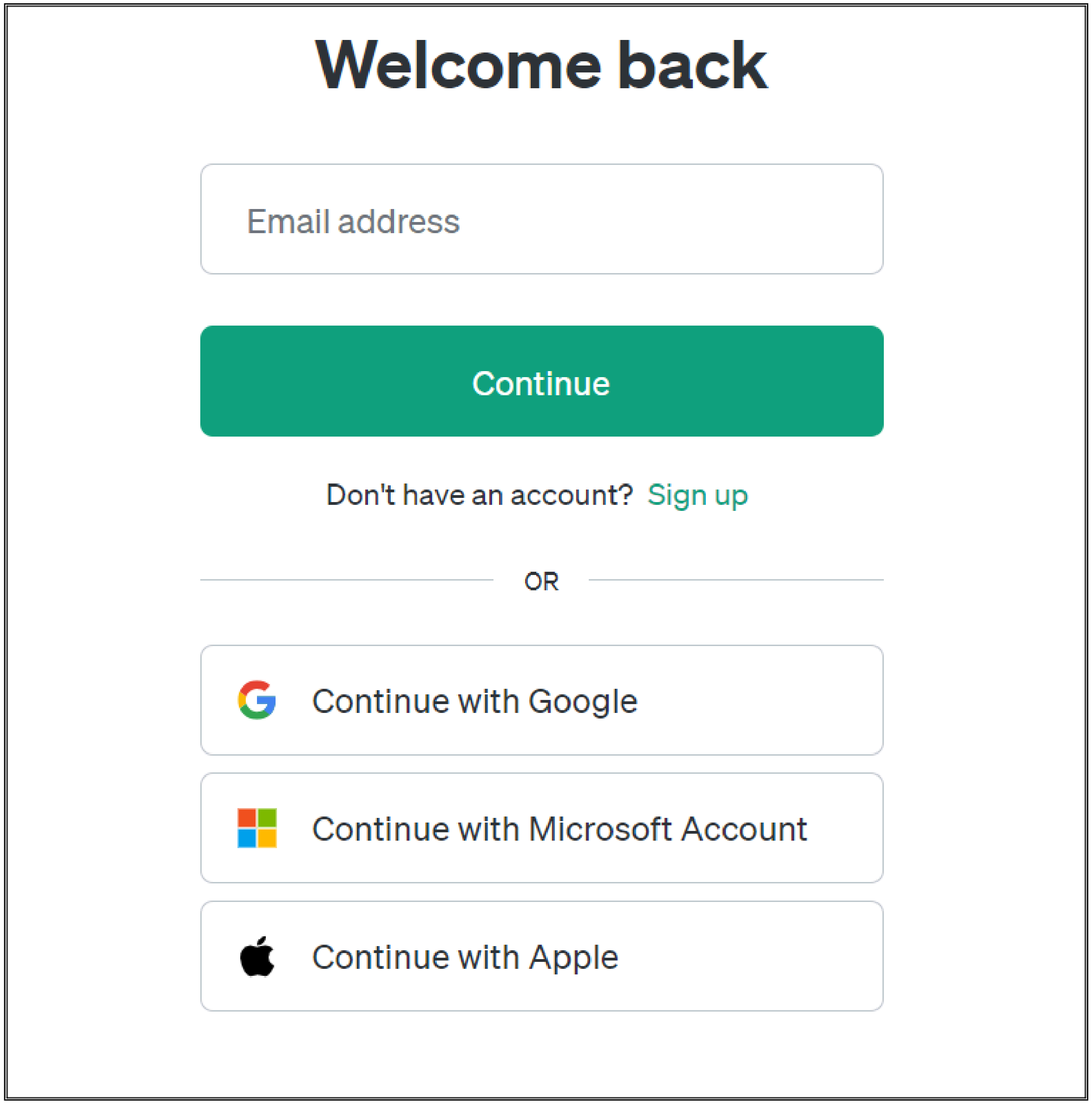
2. Once launched, you must sign up or login.

   Note: You can create a free account on ChatGPT and use GPT-3.5 features for free.

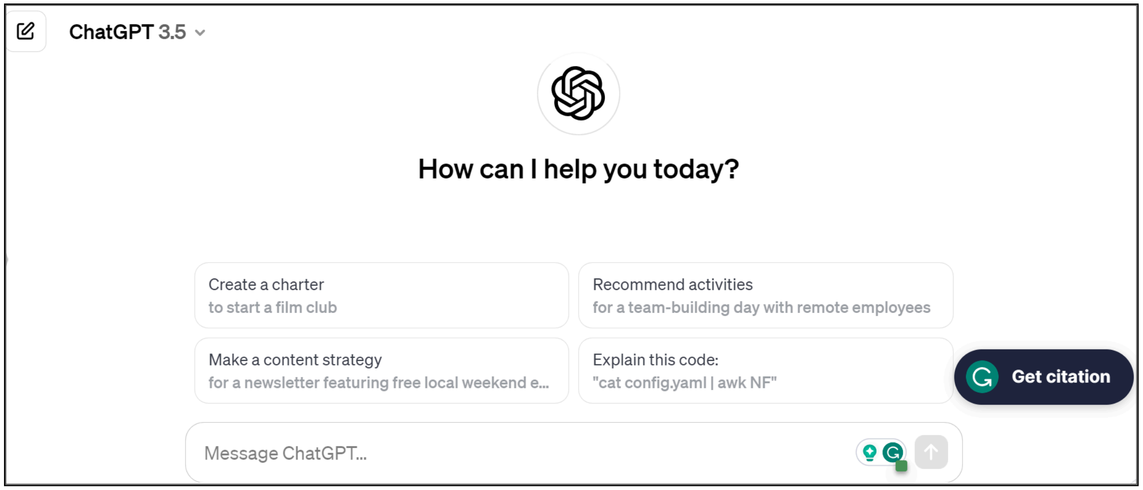


a. To sign up, you will be required to enter your email ID and password. You will receive an automated verification email. After you click verification, you will need to enter your personal details and follow the steps to complete the sign up on your first login.

b. After sign up, you will reach the login page, as shown below.



3. After login, you will view the OpenAI chatbot platform, ChatGPT, as shown below.



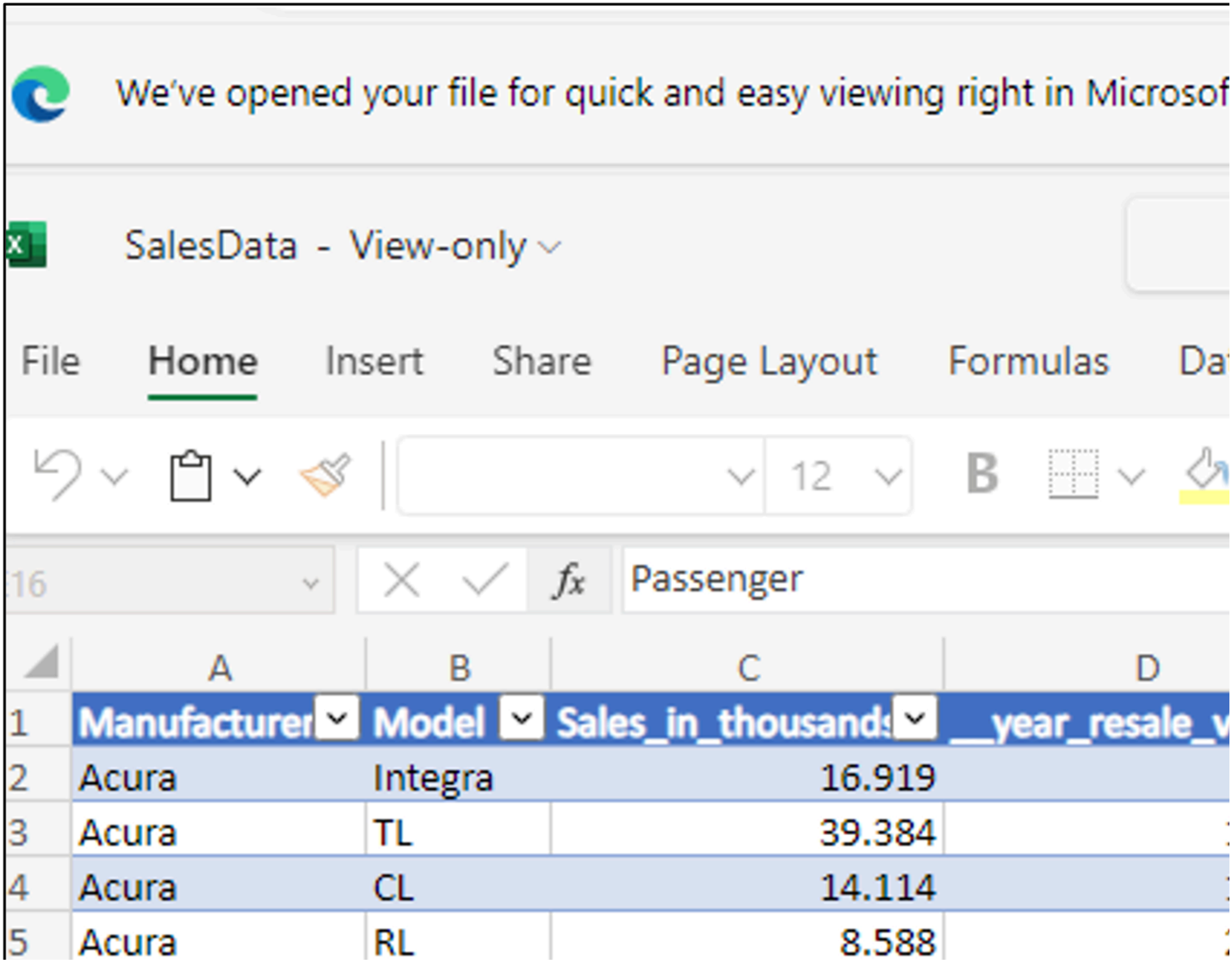
Scenario
In this lab, you'll work with an example to create dashboards for analysis of (fictitious) data for car sales. The car sales data is provided in the sheet - SalesData.

Exercise 1: Generate a sales performance dashboard

In this exercise, we will analyze monthly sales performance and engage in a conversation with ChatGPT to get insights and recommendations.

Step 1: Open the SalesData sheet in a Google Spreadsheet

1. Right-click on the link SalesData and open it in a new tab. This will download the file to your system.



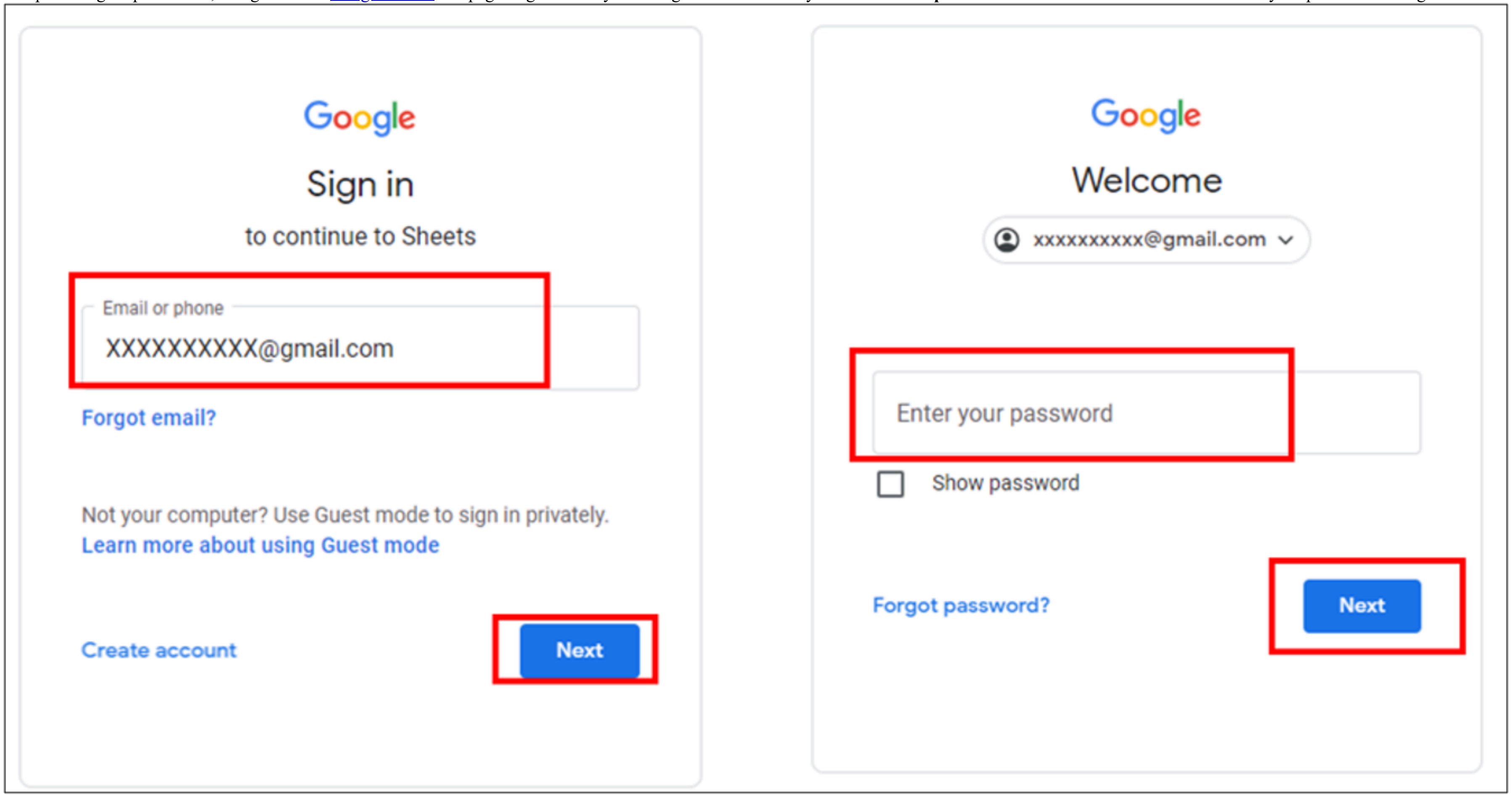
	Manufacturer	Model	Sales_in_thousands	__year_resale_v
2	Acura	Integra	16.919	
3	Acura	TL	39.384	
4	Acura	CL	14.114	
5	Acura	RL	8.588	

6	Audi	A4	20.397
7	Audi	A6	18.78
8	Audi	A8	1.38
9	BMW	323i	19.747
10	BMW	328i	9.231
11	BMW	528i	17.527
12	Buick	Century	91.561
13	Buick	Regal	39.35
14	Buick	Park Aven	27.851
15	Buick	LeSabre	83.257
16	Cadillac	DeVille	63.729
17	Cadillac	Seville	15.943
18	Cadillac	Eldorado	6.536

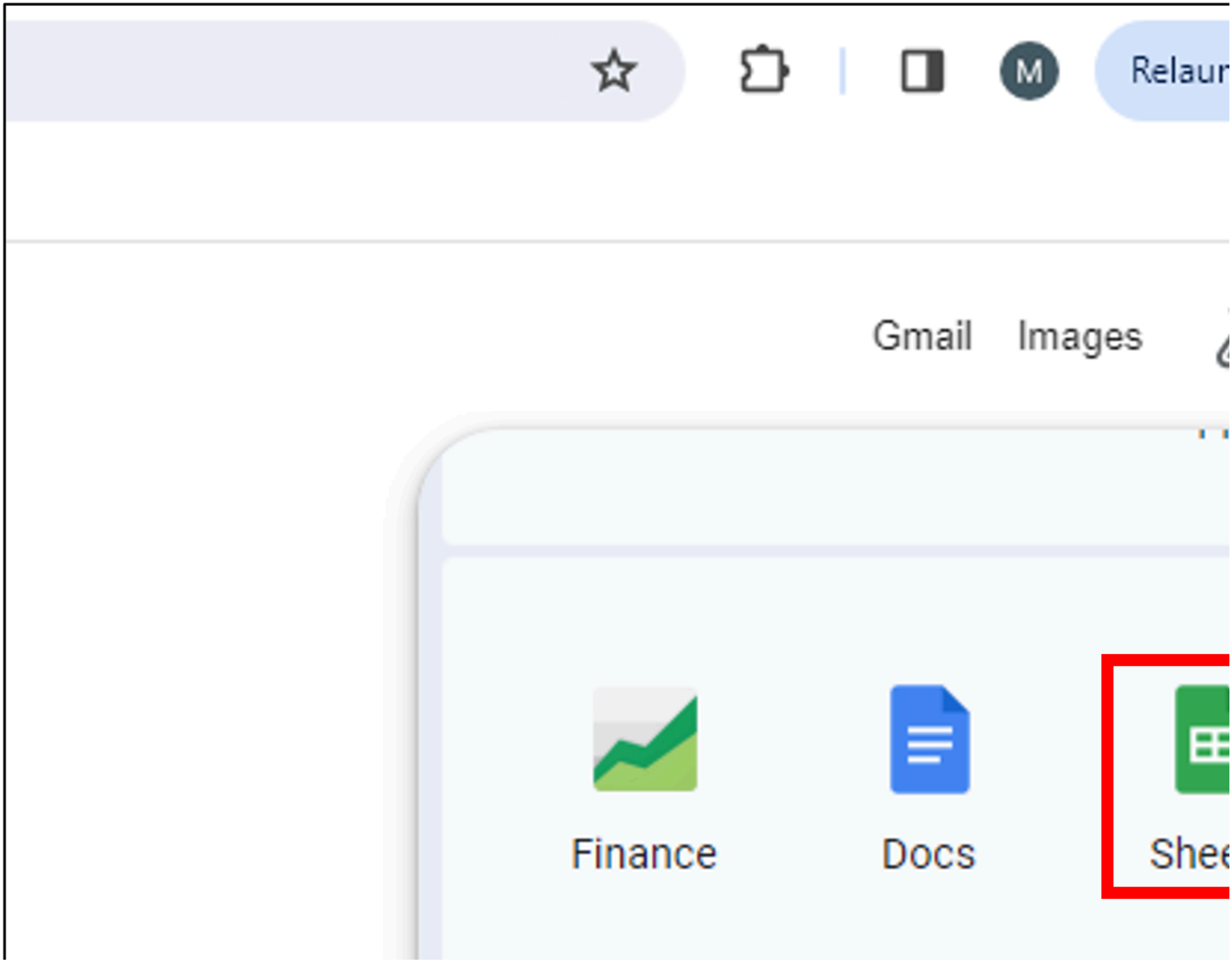
Car Sales data

Workbook Statistics

2. To open Google Speadsheets, navigate to the [Google Sheets](#) webpage. Sign in with your Google account. Enter your **email ID** or **phone number** and click on **Next**. Then enter your password and again click on **Next**.



3. Once you have logged in with your Google account, click on **Google apps** and then click on **Sheets**.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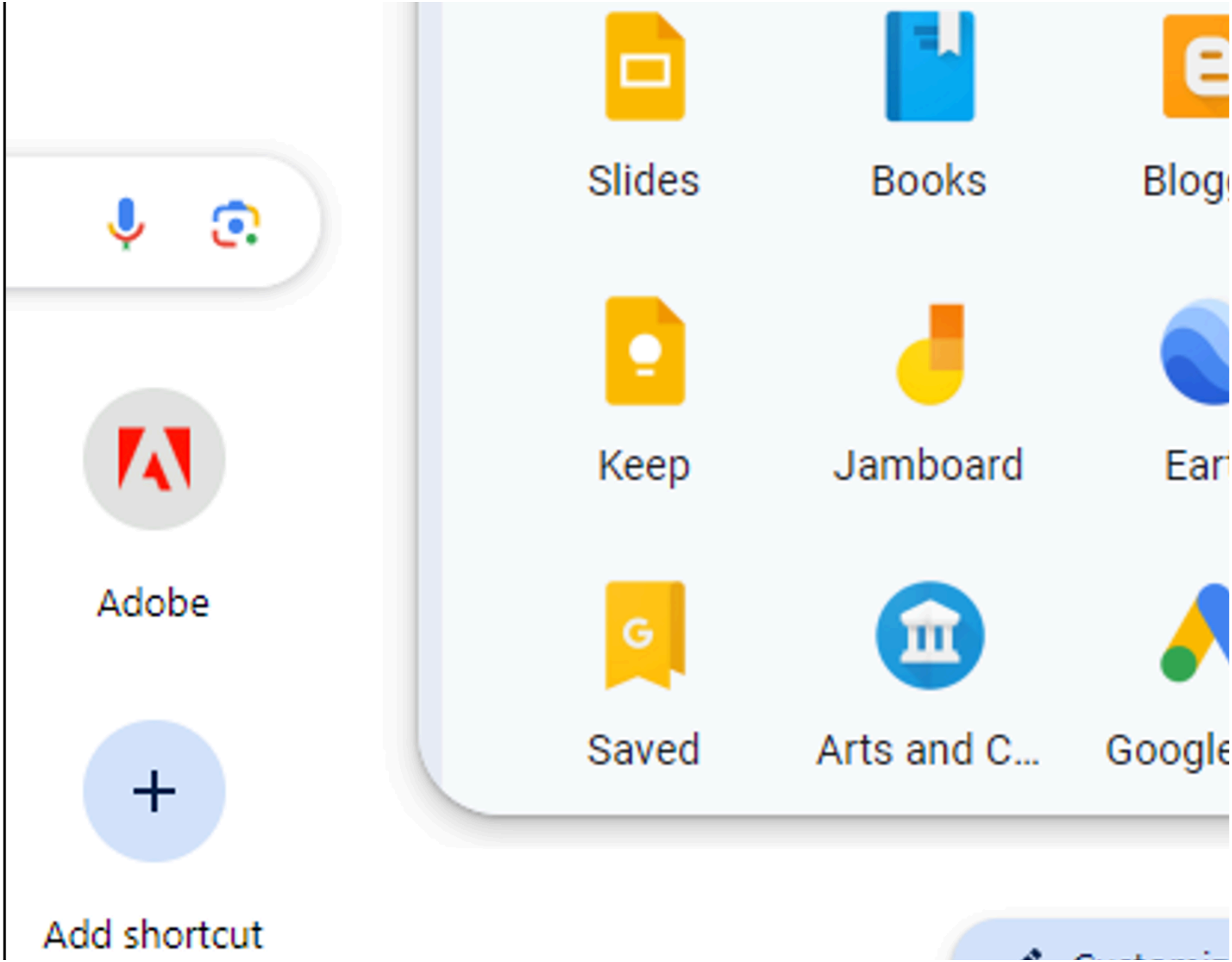
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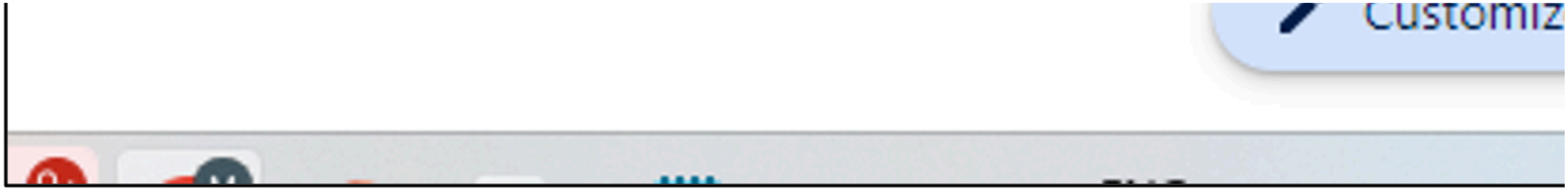
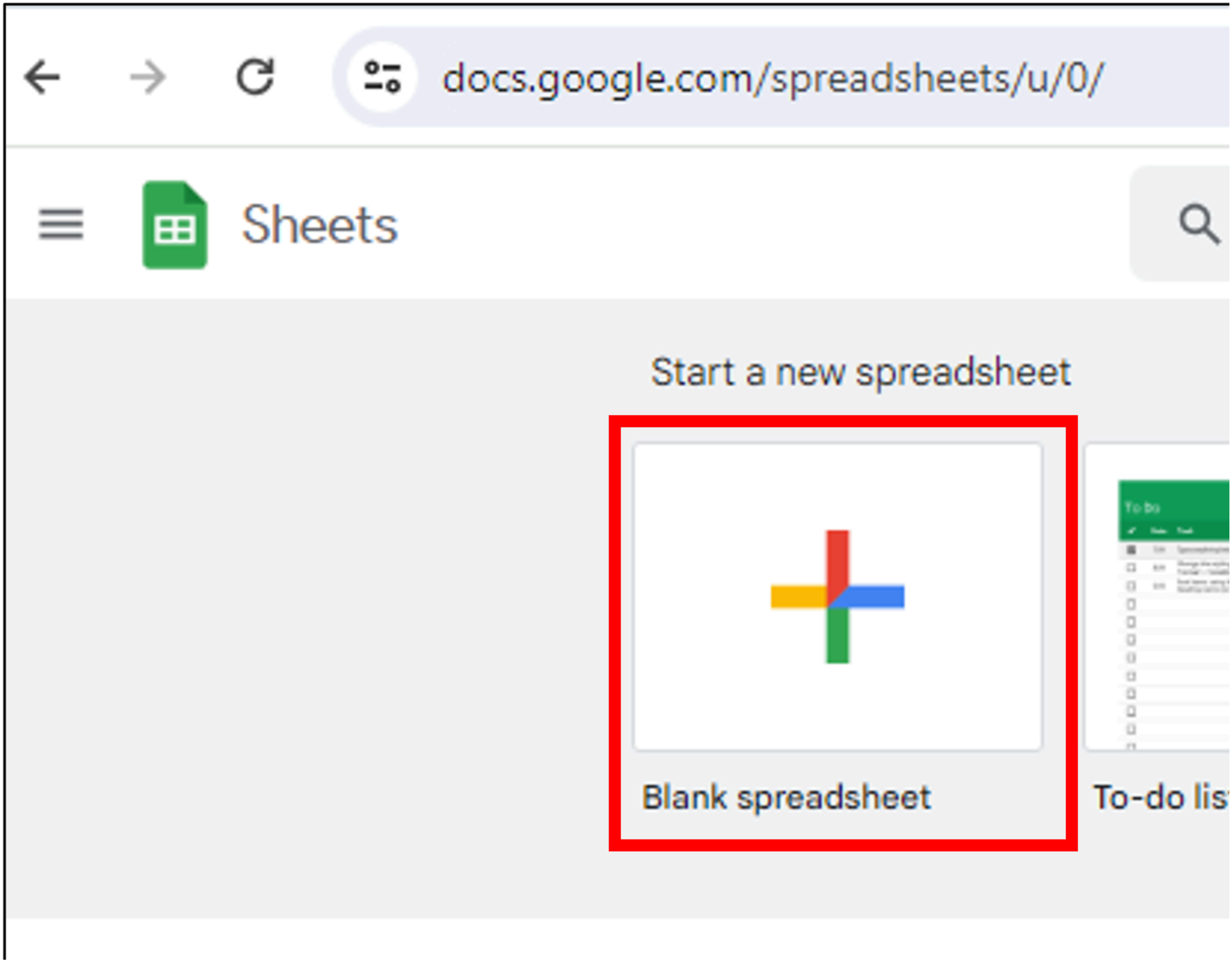


4. Click on **Blank spreadsheet**.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Sheets
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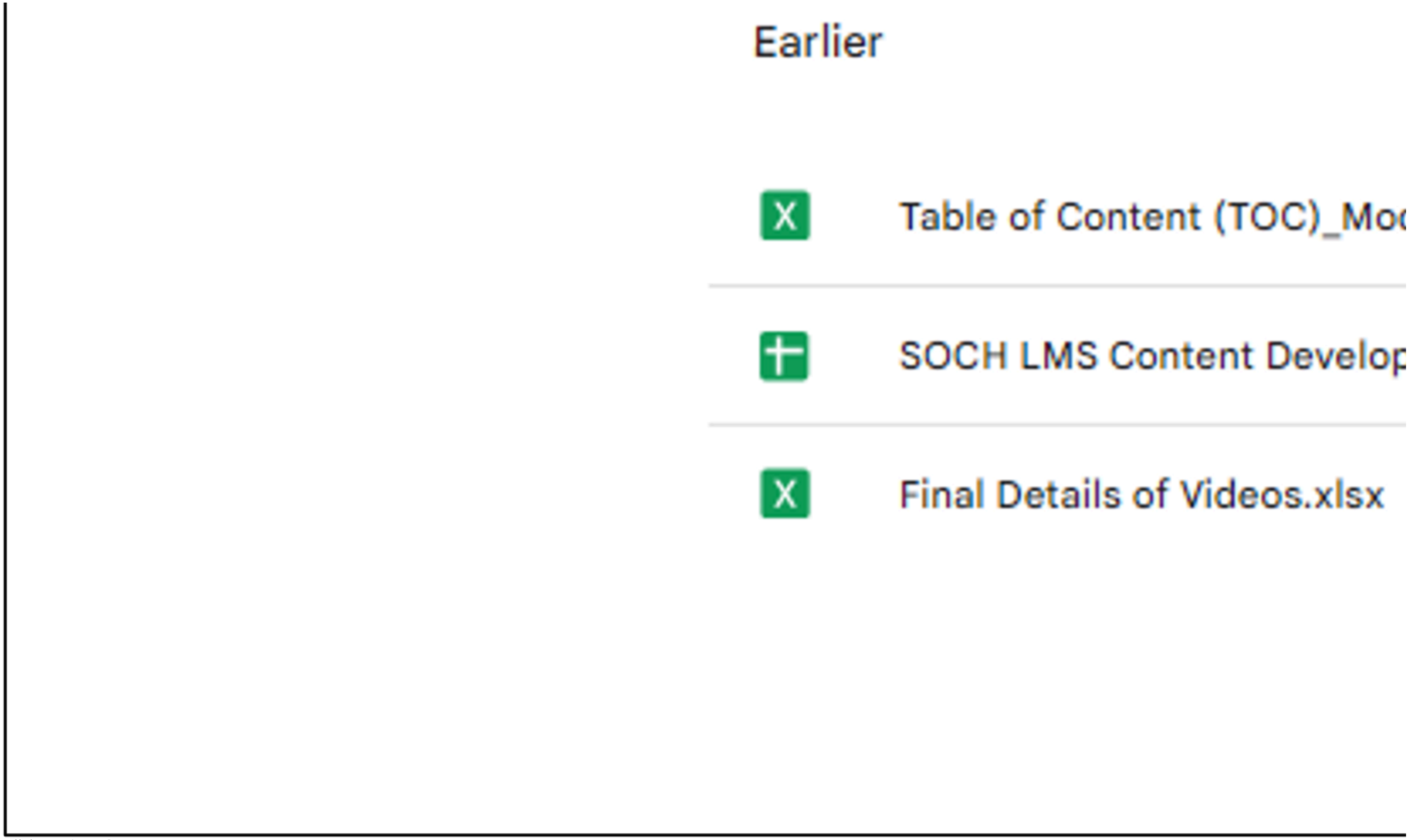
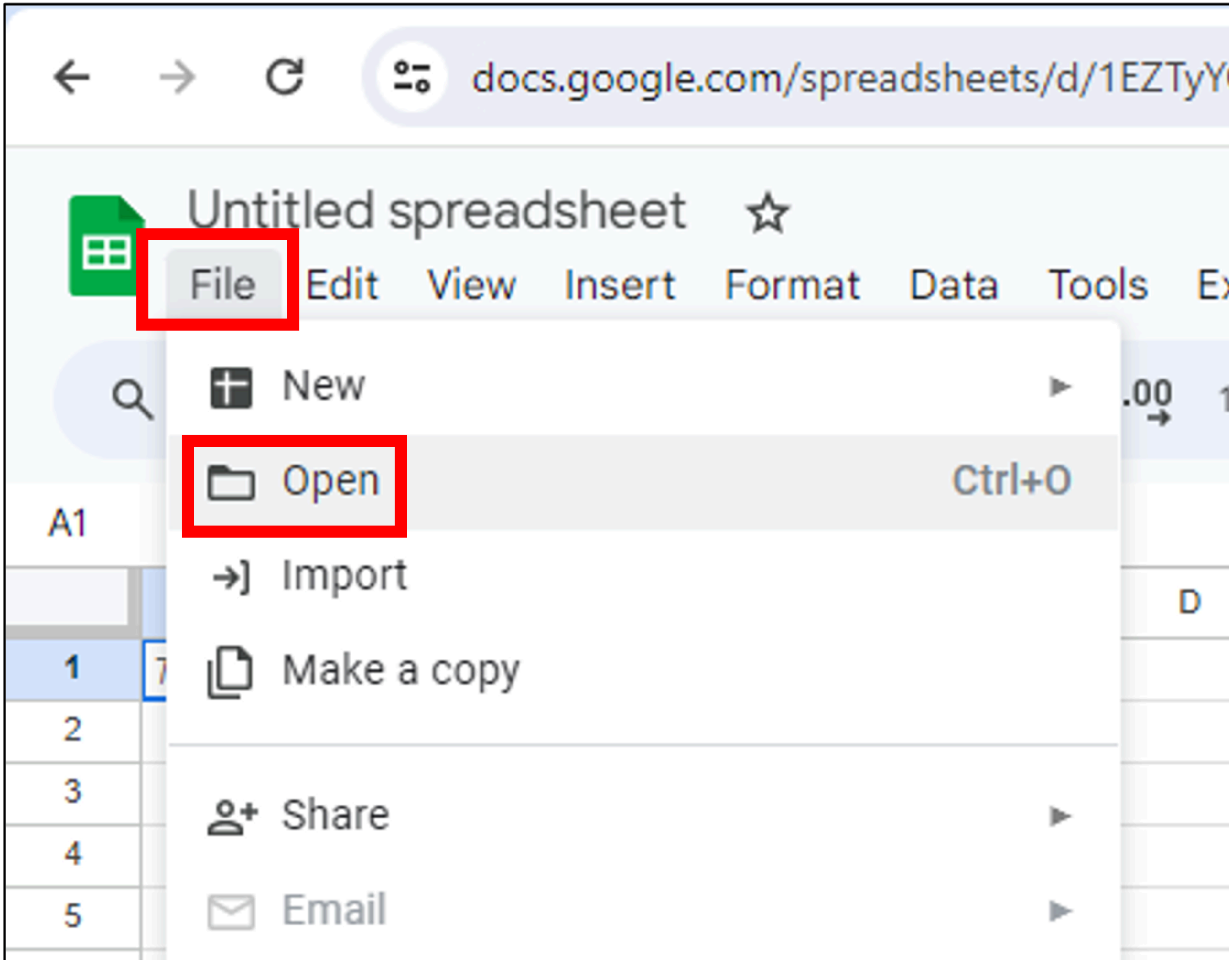


5. Click on **File** > **Open**.



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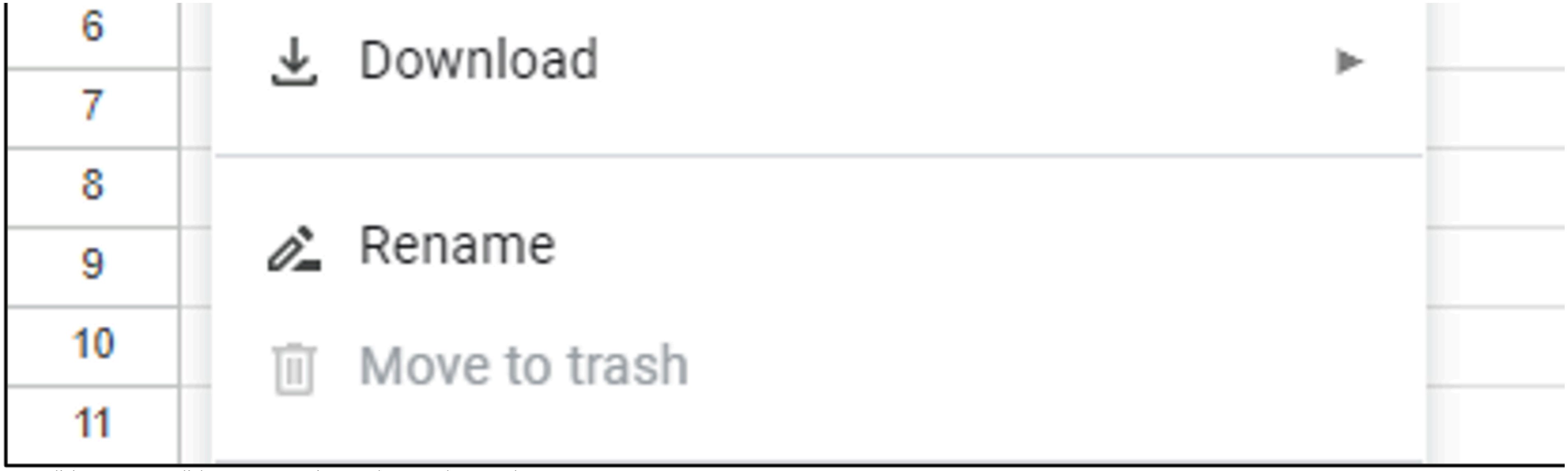
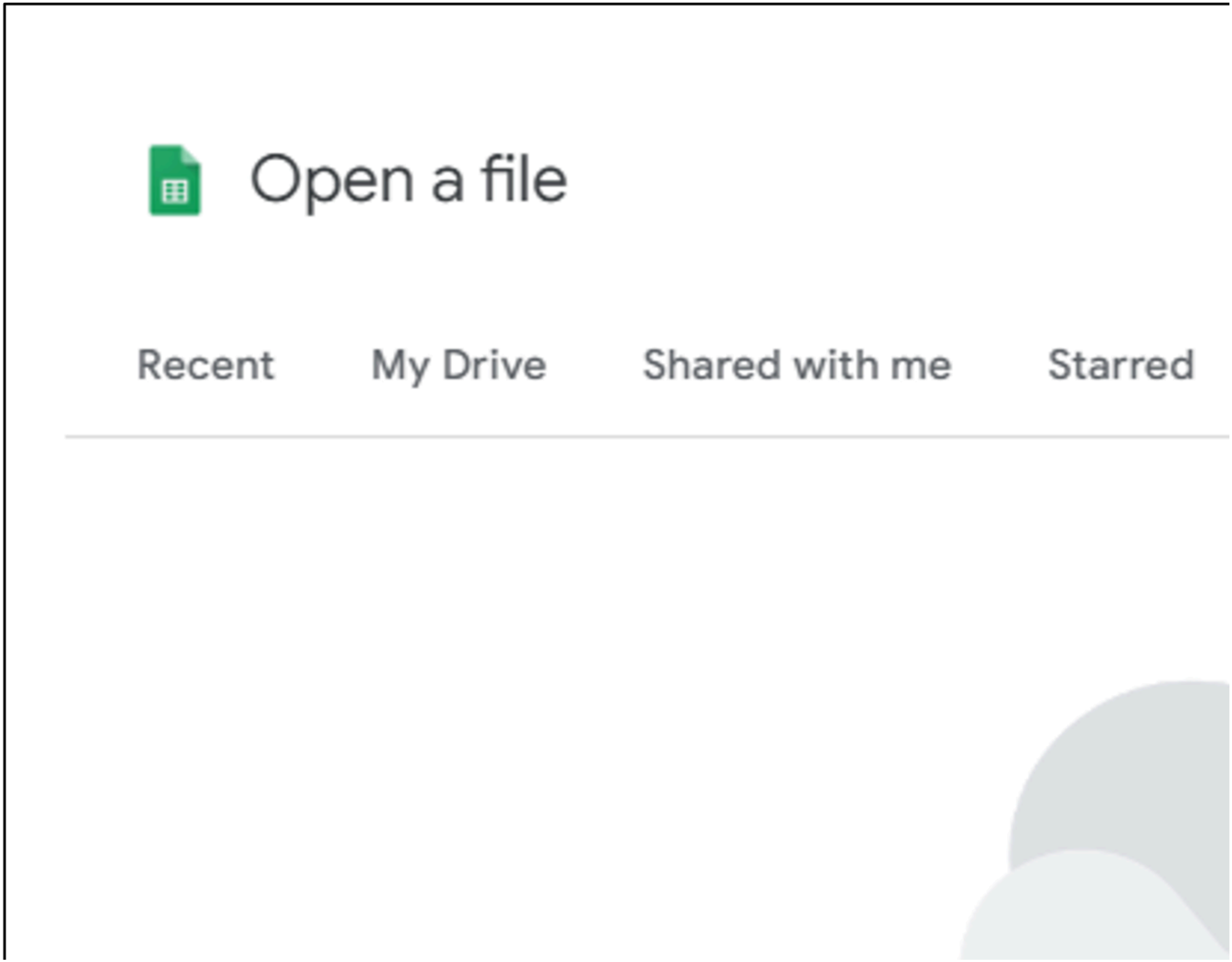


6. Next, click on **Upload**. Click **Browse** to navigate to the Car Sales Data sheet on your computer.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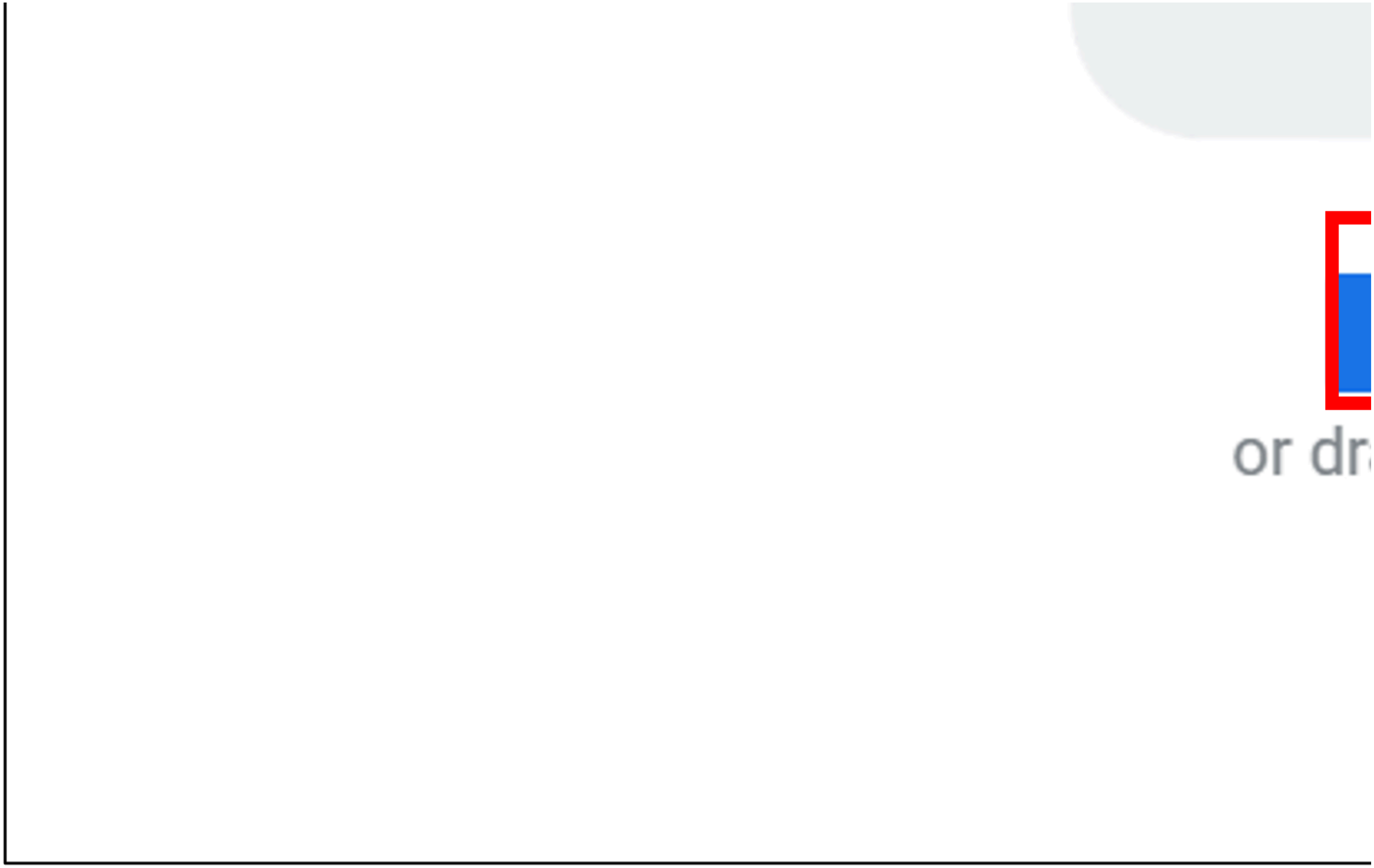
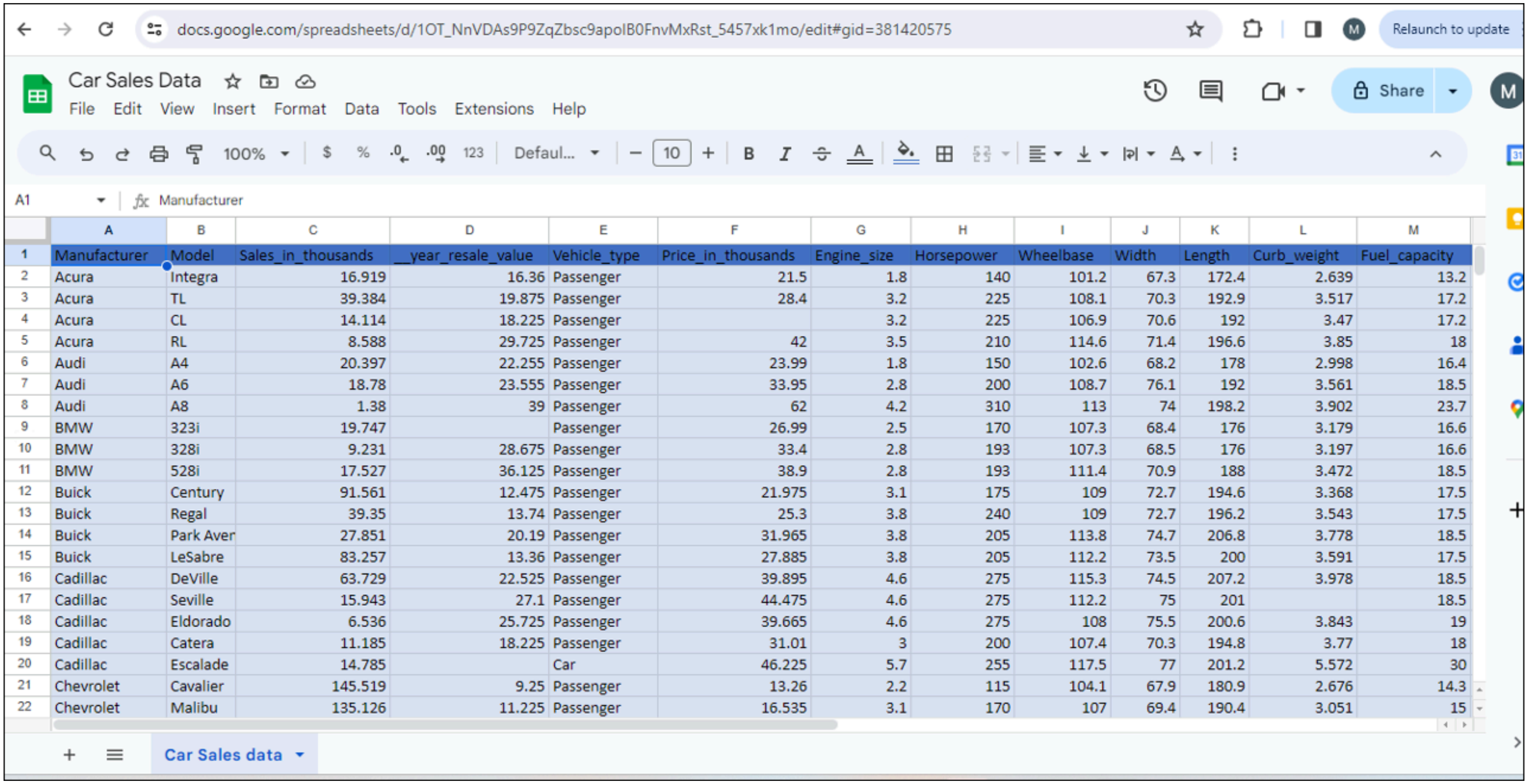


7. The sheet will open in Google Spreadsheet.



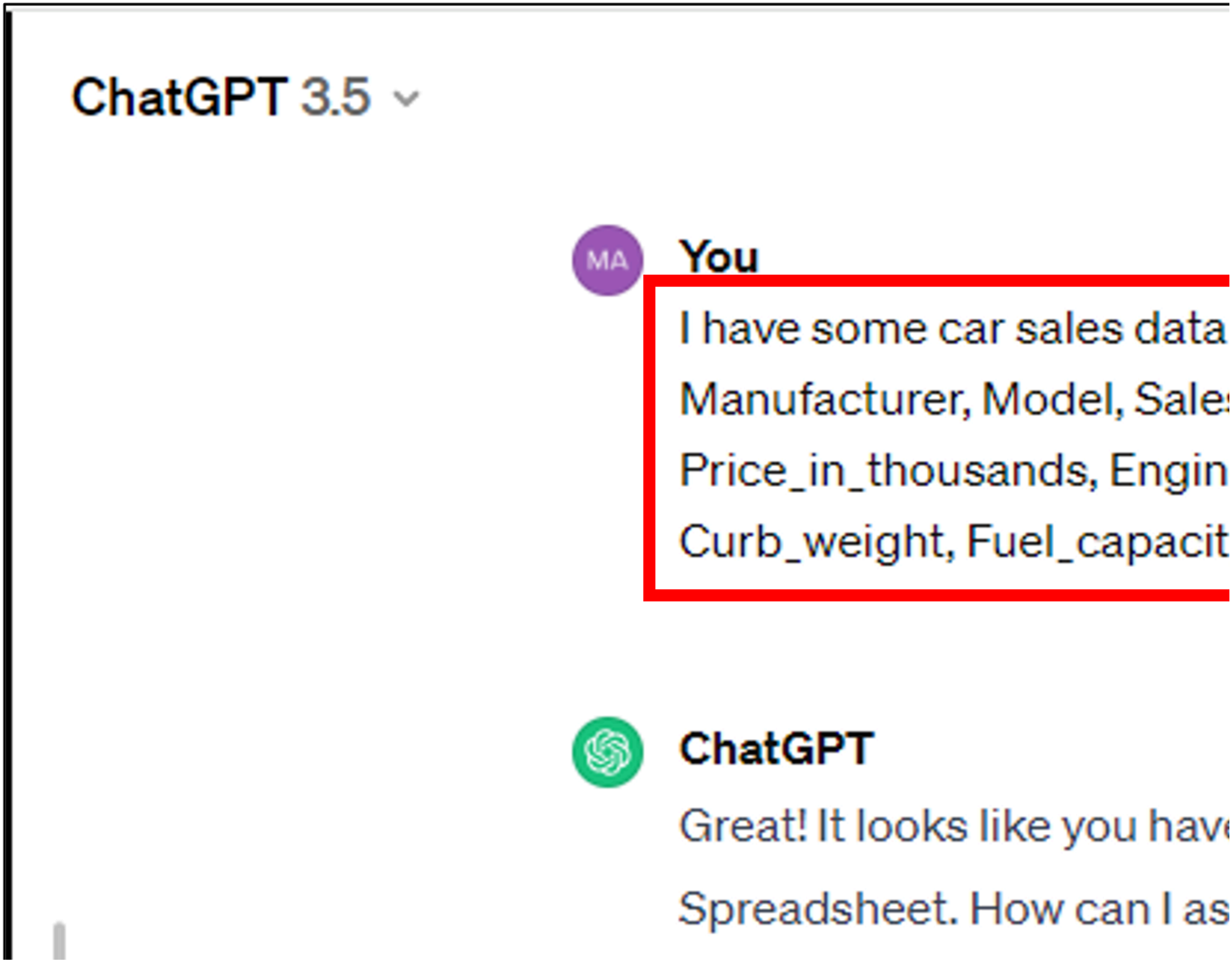
Manufacturer	Model	Sales_in_thousands	__year_resale_value	Vehicle_type	Price_in_thousands	Engine_size	Horsepower	Wheelbase	Width	Length	Curb_weight	Fuel_capacity
Acura	Integra	16.919	16.36	Passenger	21.5	1.8	140	101.2	67.3	172.4	2.639	13.2
Acura	TL	39.384	19.875	Passenger	28.4	3.2	225	108.1	70.3	192.9	3.517	17.2
Acura	CL	14.114	18.225	Passenger		3.2	225	106.9	70.6	192	3.47	17.2
Acura	RL	8.588	29.725	Passenger	42	3.5	210	114.6	71.4	196.6	3.85	18
Audi	A4	20.397	22.255	Passenger	23.99	1.8	150	102.6	68.2	178	2.998	16.4
Audi	A6	18.78	23.555	Passenger	33.95	2.8	200	108.7	76.1	192	3.561	18.5
Audi	A8	1.38	39	Passenger	62	4.2	310	113	74	198.2	3.902	23.7
BMW	323i	19.747		Passenger	26.99	2.5	170	107.3	68.4	176	3.179	16.6
BMW	328i	9.231	28.675	Passenger	33.4	2.8	193	107.3	68.5	176	3.197	16.6
BMW	528i	17.527	36.125	Passenger	38.9	2.8	193	111.4	70.9	188	3.472	18.5
Buick	Century	91.561	12.475	Passenger	21.975	3.1	175	109	72.7	194.6	3.368	17.5
Buick	Regal	39.35	13.74	Passenger	25.3	3.8	240	109	72.7	196.2	3.543	17.5
Buick	Park Aver	27.851	20.19	Passenger	31.965	3.8	205	113.8	74.7	206.8	3.778	18.5
Buick	LeSabre	83.257	13.36	Passenger	27.885	3.8	205	112.2	73.5	200	3.591	17.5
Cadillac	DeVille	63.729	22.525	Passenger	39.895	4.6	275	115.3	74.5	207.2	3.978	18.5
Cadillac	Seville	15.943	27.1	Passenger	44.475	4.6	275	112.2	75	201		18.5
Cadillac	Eldorado	6.536	25.725	Passenger	39.665	4.6	275	108	75.5	200.6	3.843	19
Cadillac	Catera	11.185	18.225	Passenger	31.01	3	200	107.4	70.3	194.8	3.77	18
Cadillac	Escalade	14.785		Car	46.225	5.7	255	117.5	77	201.2	5.572	30
Chevrolet	Cavalier	145.519	9.25	Passenger	13.26	2.2	115	104.1	67.9	180.9	2.676	14.3
Chevrolet	Malibu	135.126	11.225	Passenger	16.535	3.1	170	107	69.4	190.4	3.051	15

Step 2: Identify generic approach for creating a dashboard

2. Let's consider prompting ChatGPT to create a dashboard for the car sales data. Enter the following prompt.

```
I have some car sales data in Google spreadsheet with the following columns – Manufacturer, Model, Sales_in_thousands, __year_resale_value, Vehicle_type, Price_in_thousands, Engine_size, Horsepower,
```

3. After you enter the above prompt, ChatGPT will ask for more details for better understanding.


ChatGPT 3.5
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I have some car sales data
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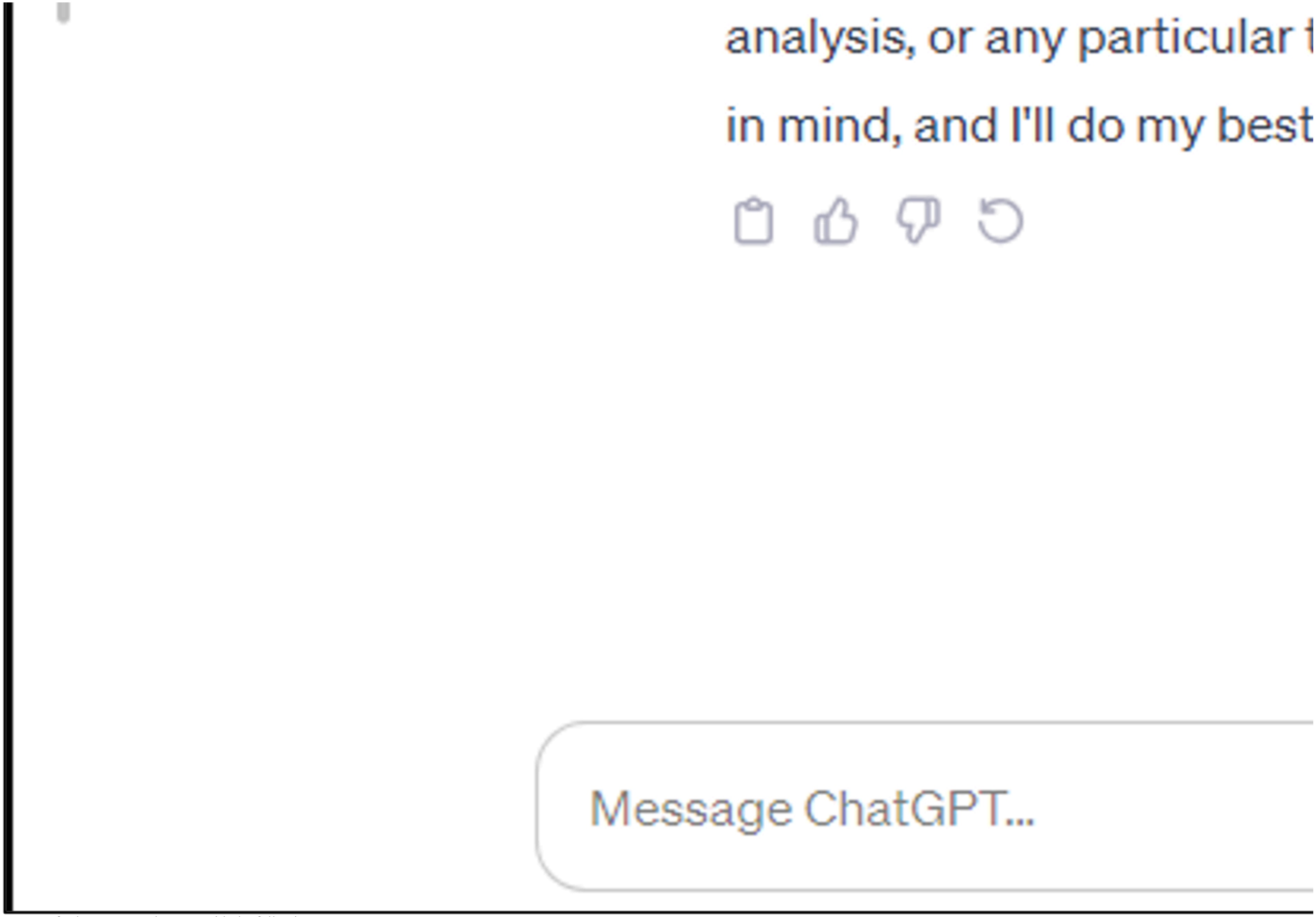
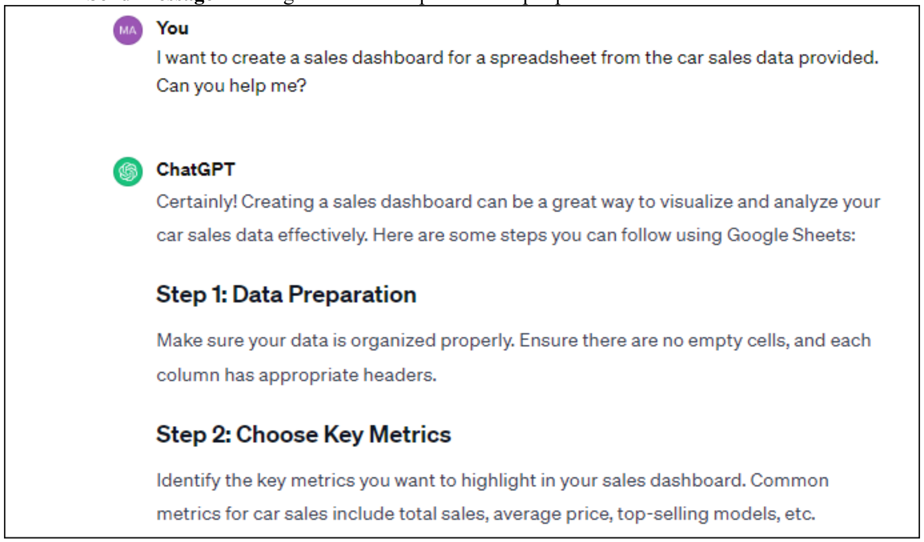


4. You can further prompt ChatGPT with the following statement.

```
I want to create a sales dashboard for a spreadsheet from the car sales data provided. Can you help me?
```

5. Click the **Send Message** icon to generate the output. The output produced at our end is shown here. In the output, ChatGPT has provided a stepwise approach to creating a dashboard.



You

I want to create a sales dashboard for a spreadsheet from the car sales data provided. Can you help me?

ChatGPT

Certainly! Creating a sales dashboard can be a great way to visualize and analyze your car sales data effectively. Here are some steps you can follow using Google Sheets:

Step 1: Data Preparation

Make sure your data is organized properly. Ensure there are no empty cells, and each column has appropriate headers.

Step 2: Choose Key Metrics

Identify the key metrics you want to highlight in your sales dashboard. Common metrics for car sales include total sales, average price, top-selling models, etc.

Step 3: Create Pivot Tables

Use pivot tables to summarize and analyze your data. You can create pivot tables for total sales, average price, or any other metrics you've identified.

1. **Insert Pivot Table:**
   - Highlight your data.
   - Go to `Data > Pivot table`.
   - Choose where to place the pivot table.
2. **Configure Pivot Table:**
   - Drag the fields you need (e.g., Month, Manufacturer, Model) to Rows and Values.
   - Choose the aggregation method for the values (e.g., sum for total sales).

Step 4: Create Charts

Visualize your data with charts. Google Sheets offers various chart types such as bar charts, line charts, and pie charts.

1. **Insert Chart:**
   - Select the data you want to include in the chart.
   - Go to `Insert > Chart`.
2. **Configure Chart:**
   - Choose the chart type.
   - Adjust the chart settings, titles, and legends.

Step 5: Combine Charts into a Dashboard

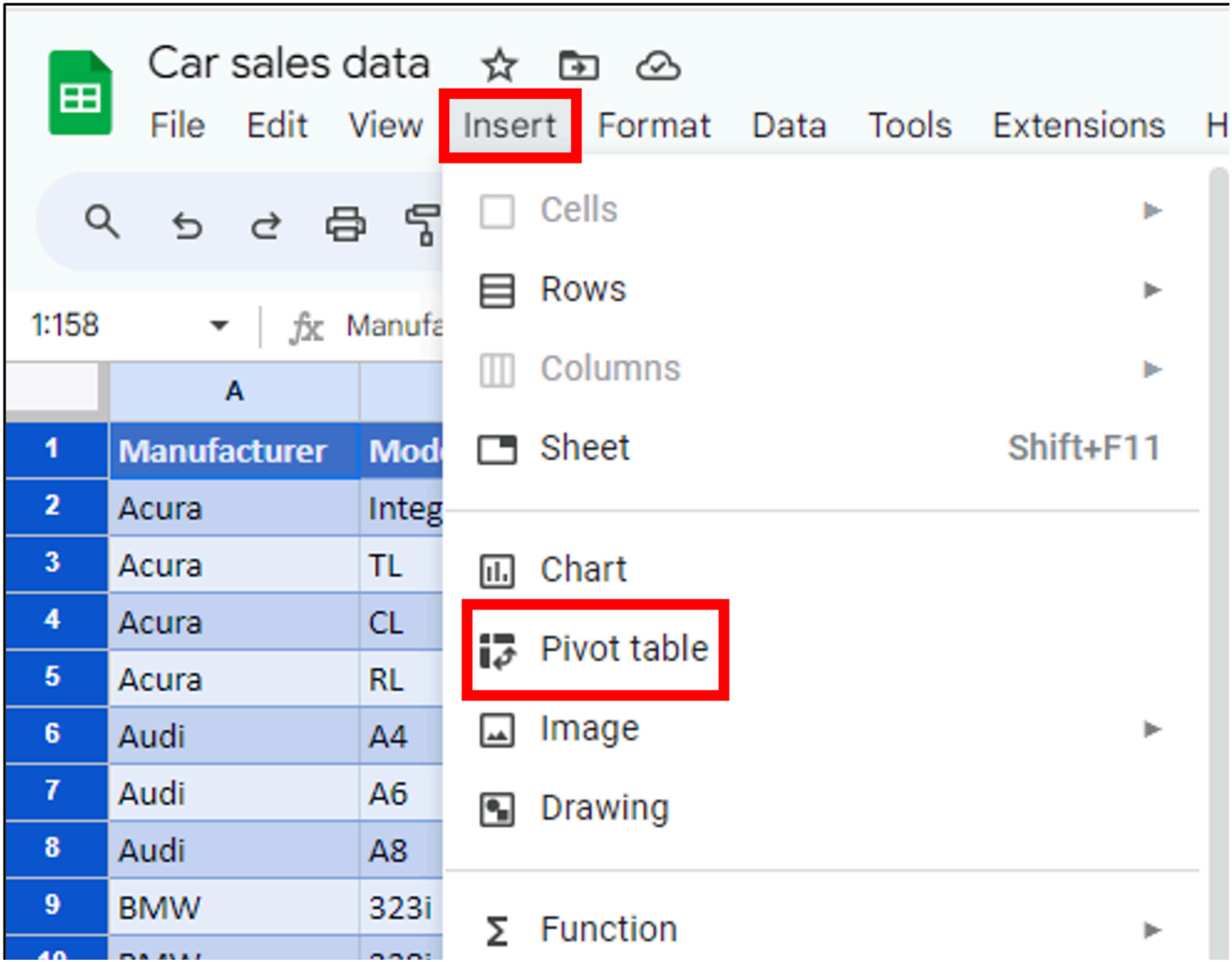
You can create a separate sheet for your dashboard and arrange your charts to provide a comprehensive view of your data.

1. **Insert New Sheet:**
   - Click on the `+` button at the bottom to add a new sheet.
2. **Copy and Paste Charts:**
   - Copy the charts from the individual sheets and paste them into your dashboard sheet.
3. **Organize the Dashboard:**
   - Arrange the charts in a way that makes sense for your analysis.

Step 3: Create Pivot Tables

Considering the stepwise approach suggested by ChatGPT, we first need to create the pivot tables to summarize and analyze the data.

1. Pivot tables allow you to summarize and analyze your data quickly. Let's create a pivot table to gather insights. Open the spreadsheet named "**Car sales data**". Select the entire data, click on **Insert**, and then click on **Pivot table**.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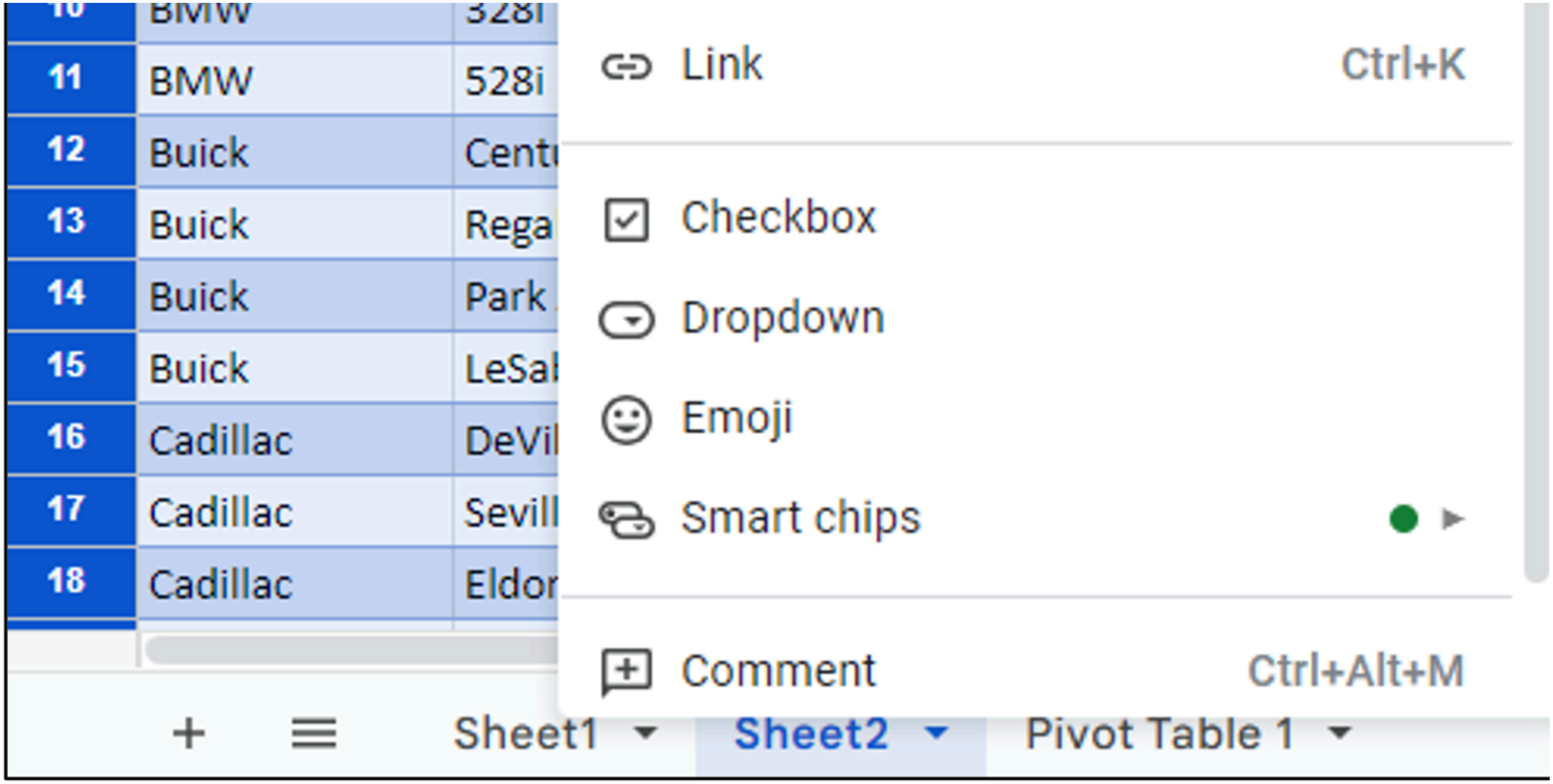
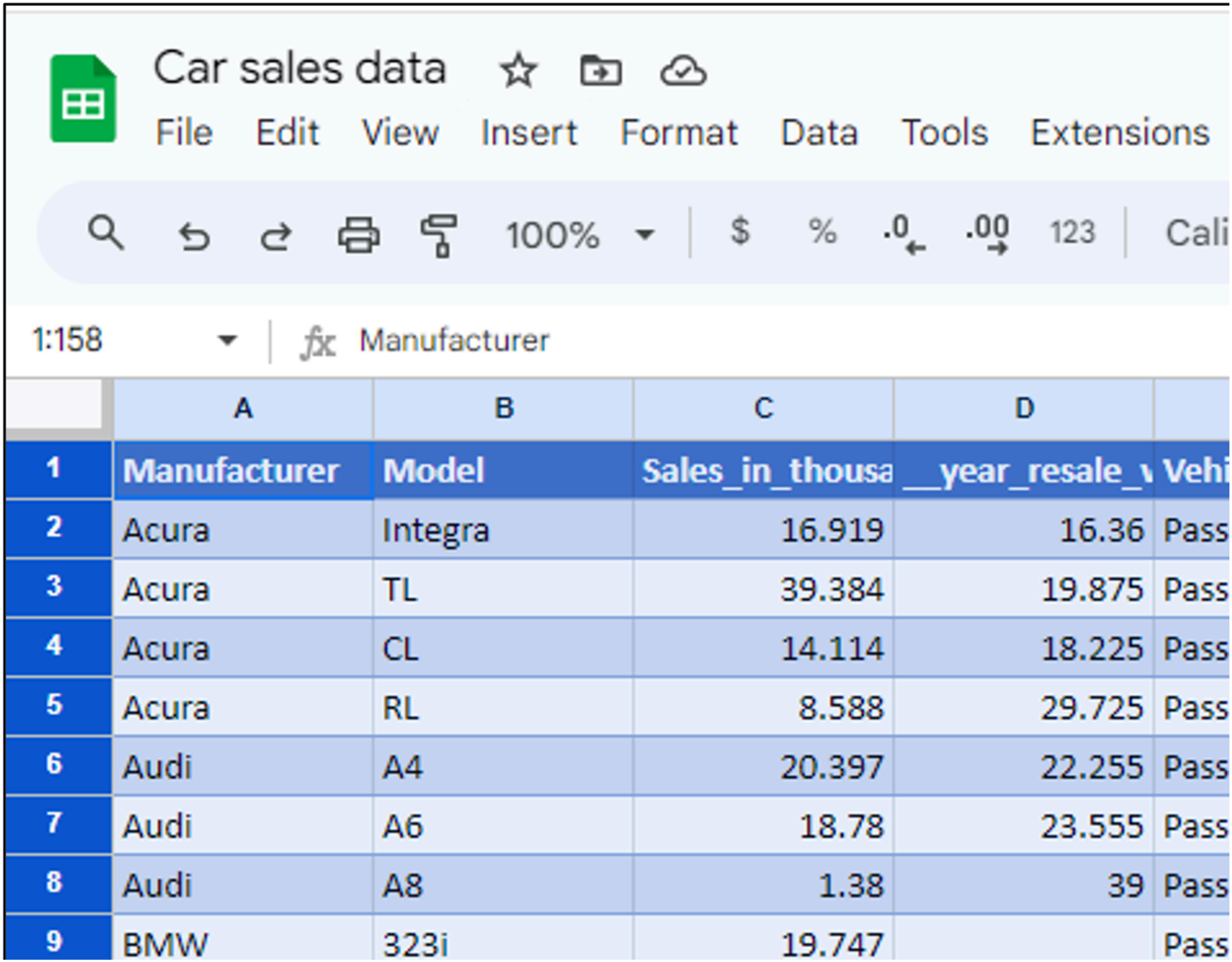


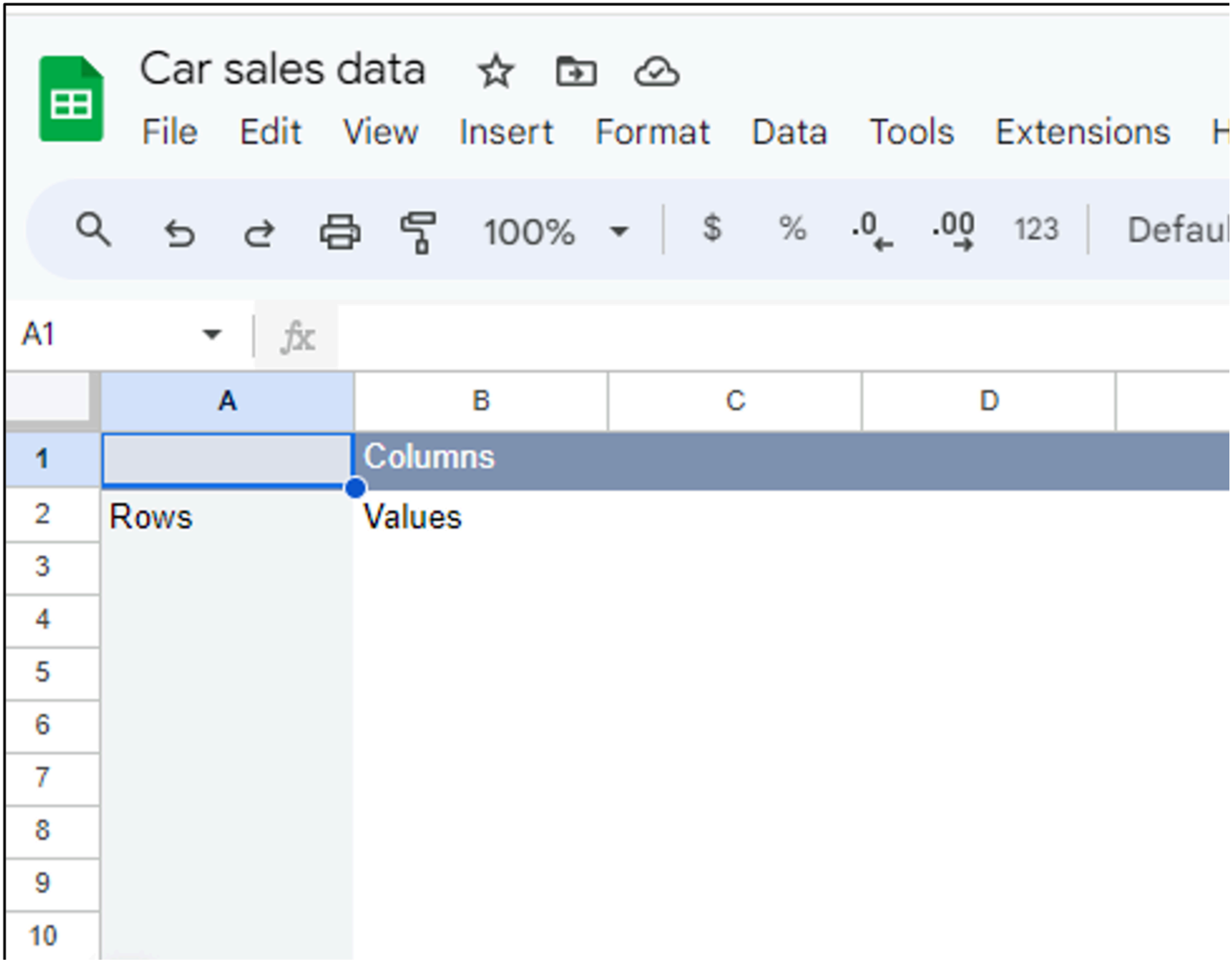
2. Click on **Create** to start creating the pivot table.



	A	B	C	D	
1	Manufacturer	Model	Sales_in_thousa	__year_resale_v	Vehi
2	Acura	Integra	16.919	16.36	Pass
3	Acura	TL	39.384	19.875	Pass
4	Acura	CL	14.114	18.225	Pass
5	Acura	RL	8.588	29.725	Pass
6	Audi	A4	20.397	22.255	Pass
7	Audi	A6	18.78	23.555	Pass
8	Audi	A8	1.38	39	Pass
9	BMW	323i	19.747		Pass

10	BMW	328i	9.231	28.675	Pass
11	BMW	528i	17.527	36.125	Pass
12	Buick	Century	91.561	12.475	Pass
13	Buick	Regal	39.35	13.74	Pass
14	Buick	Park Avenue	27.851	20.19	Pass
15	Buick	LeSabre	83.257	13.36	Pass
16	Cadillac	DeVille	63.729	22.525	Pass
17	Cadillac	Seville	15.943	27.1	Pass
18	Cadillac	Eldorado	6.536	25.725	Pass

3. Drag **Manufacturer** to **Rows** and **Sales_in_thousands** to the **Values** field.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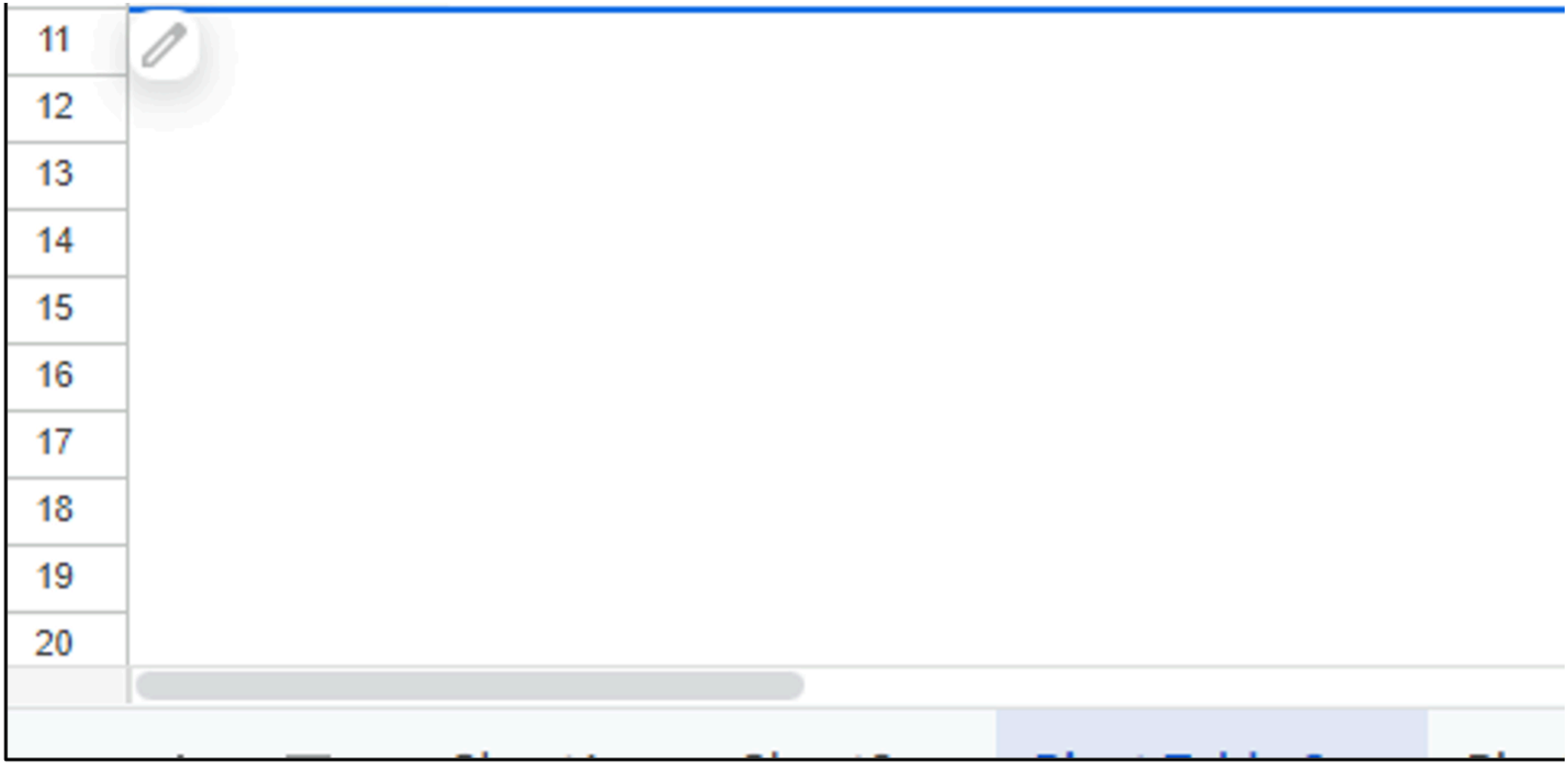
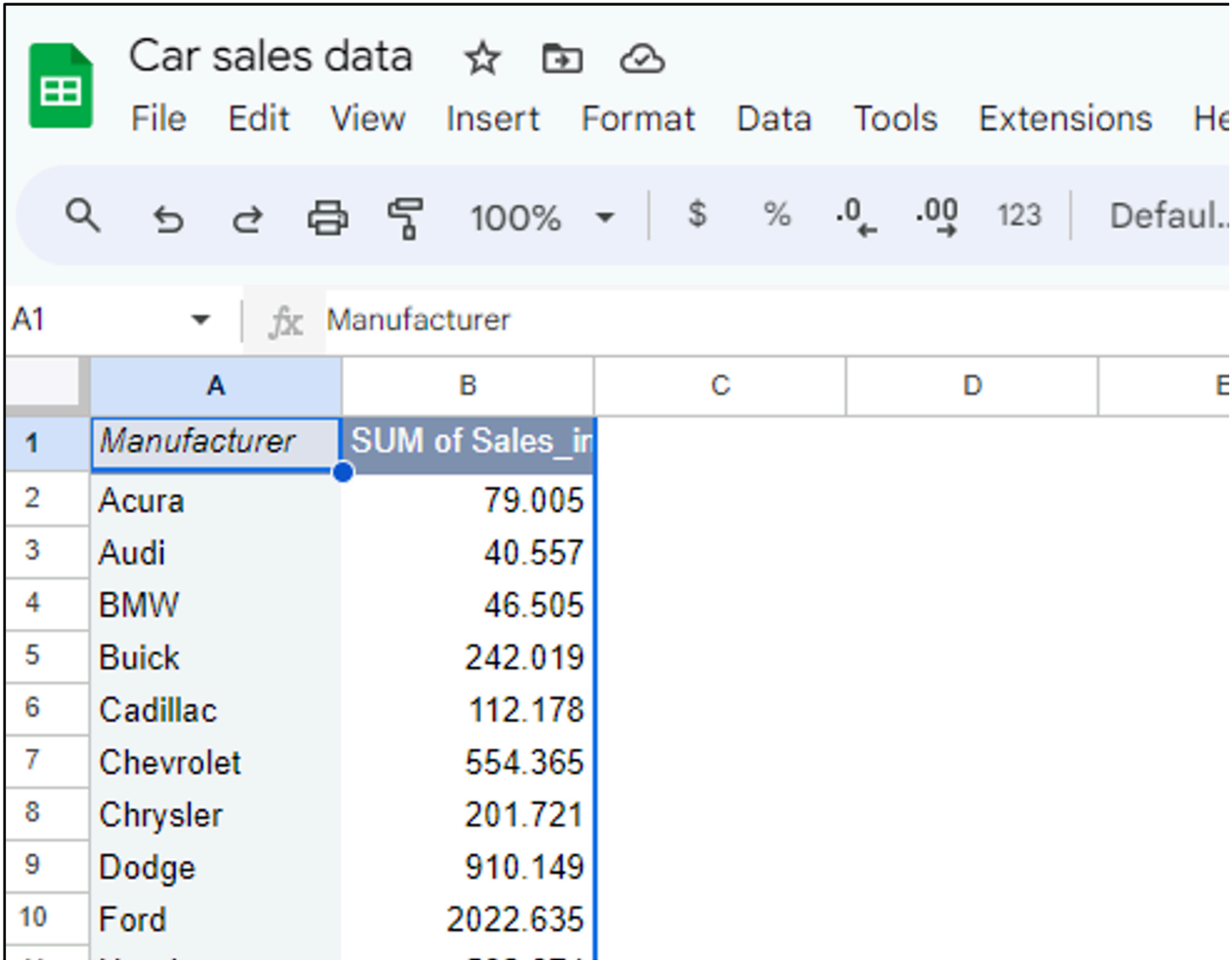
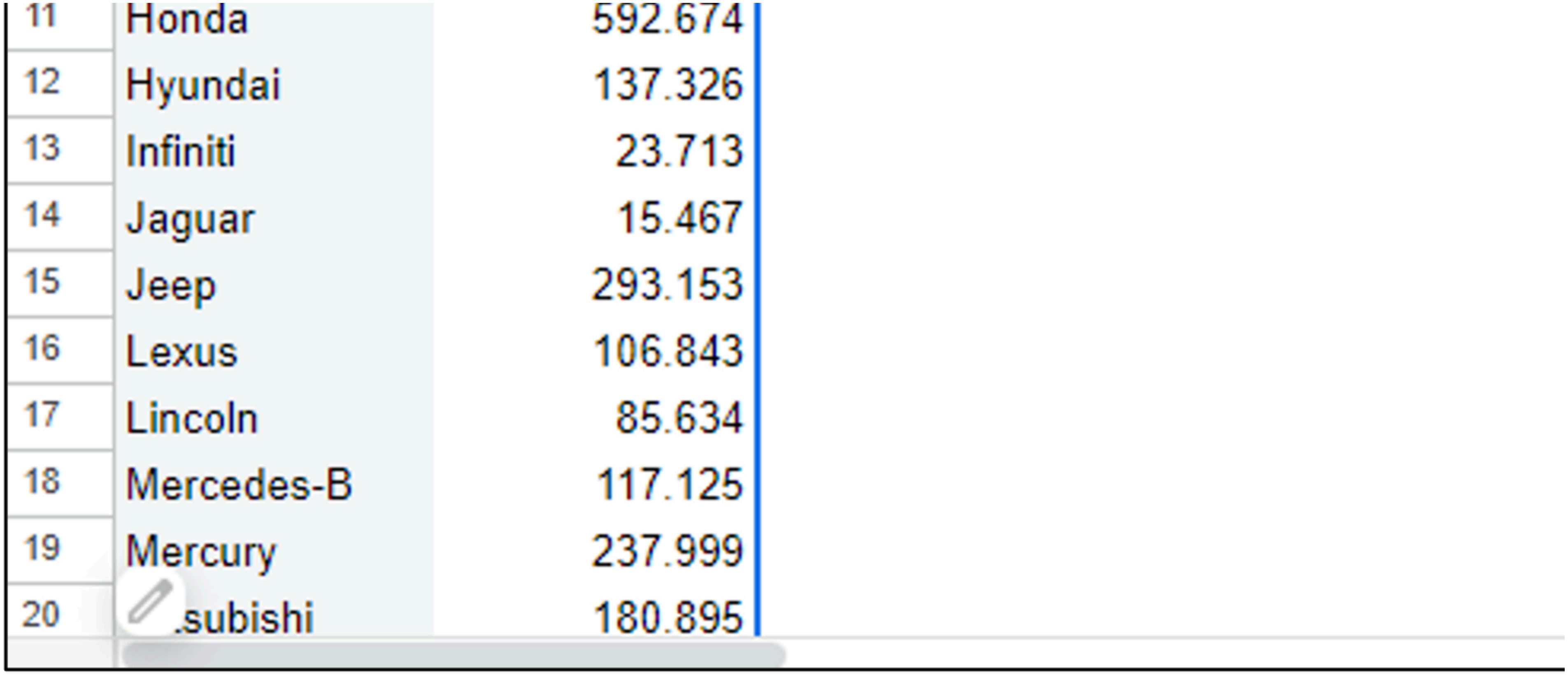


4. Click on the arrow under the **Values** field and select **SUM**.

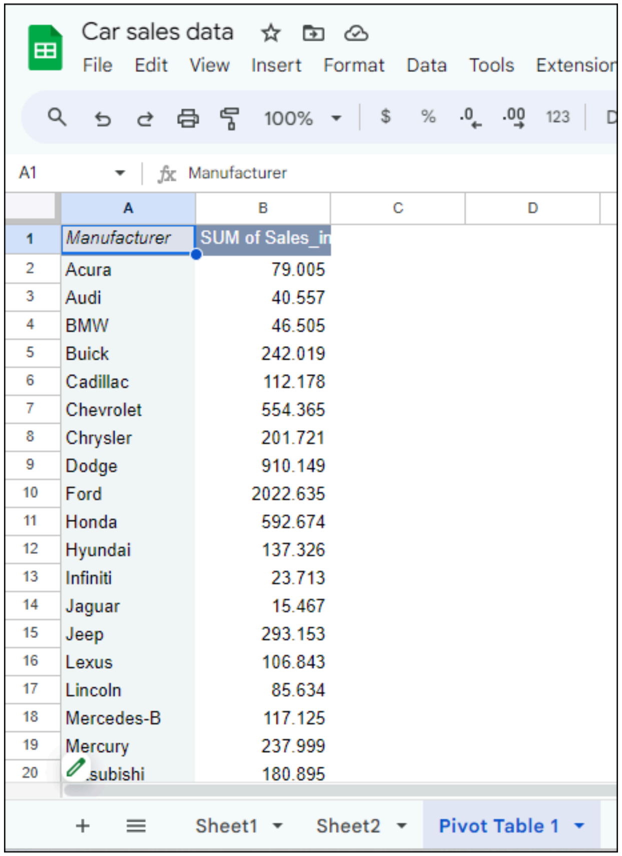


	A	B
1	*Manufacturer*	SUM of Sales_in
2	Acura	79.005
3	Audi	40.557
4	BMW	46.505
5	Buick	242.019
6	Cadillac	112.178
7	Chevrolet	554.365
8	Chrysler	201.721
9	Dodge	910.149
10	Ford	2022.635

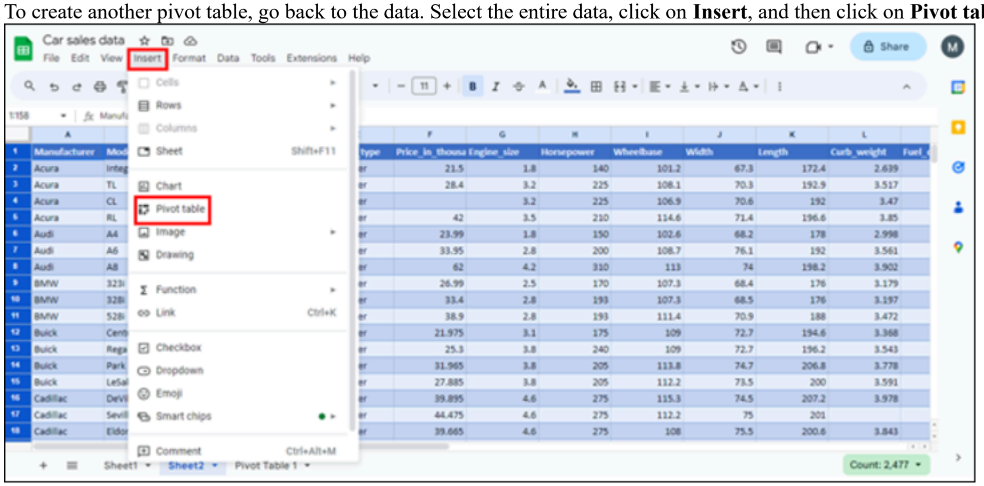


11	Honda	592.674
12	Hyundai	137.326
13	Infiniti	23.713
14	Jaguar	15.467
15	Jeep	293.153
16	Lexus	106.843
17	Lincoln	85.634
18	Mercedes-B	117.125
19	Mercury	237.999
20	[illegible]subishi	180.895

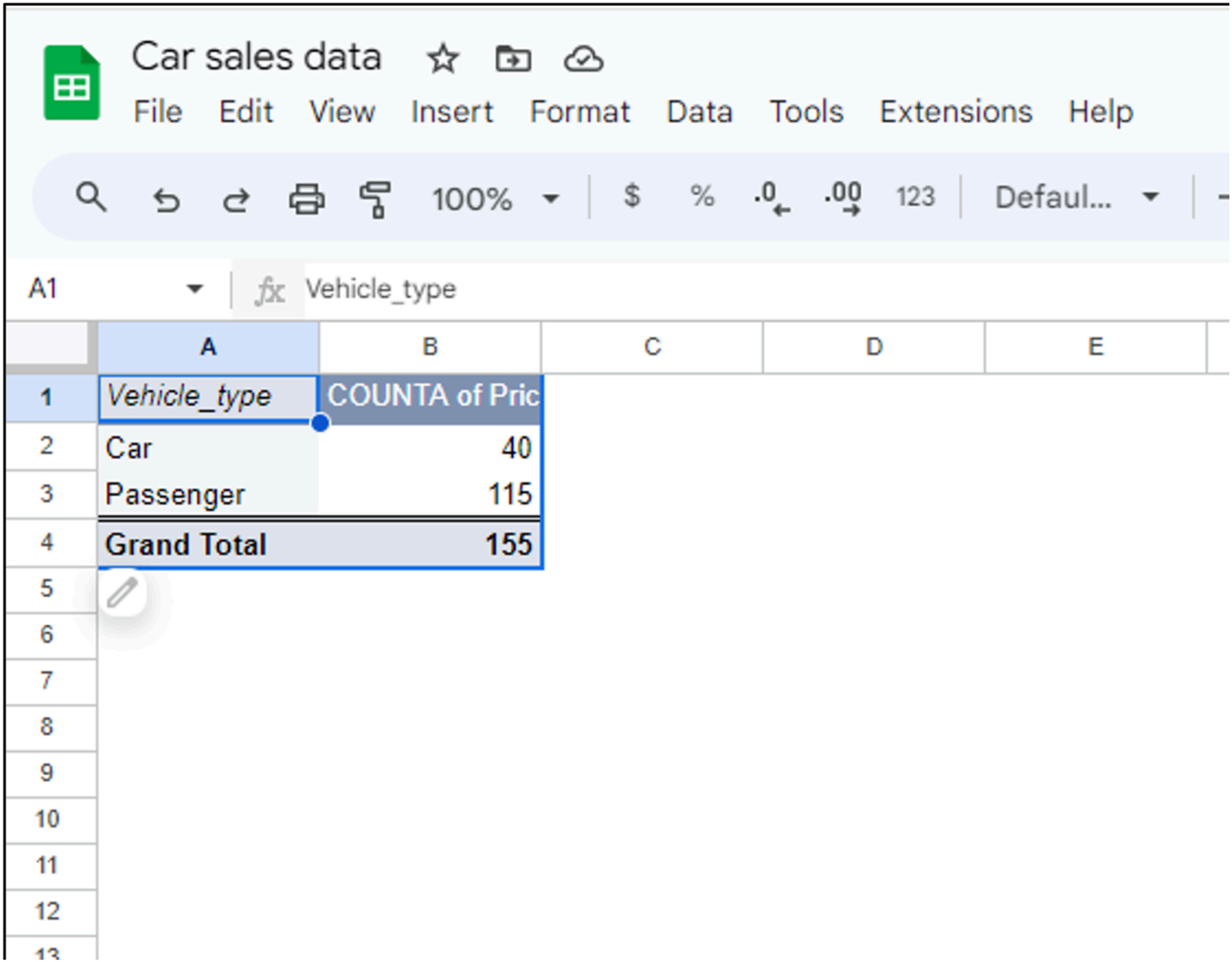
5. The pivot table is ready. You can also change the name of this table.



6. To create another pivot table, go back to the data. Select the entire data, click on **Insert**, and then click on **Pivot table**.



7. This time, drag **Vehicle_type** to **Rows** and **Price_in_thousands** to the **Values** field.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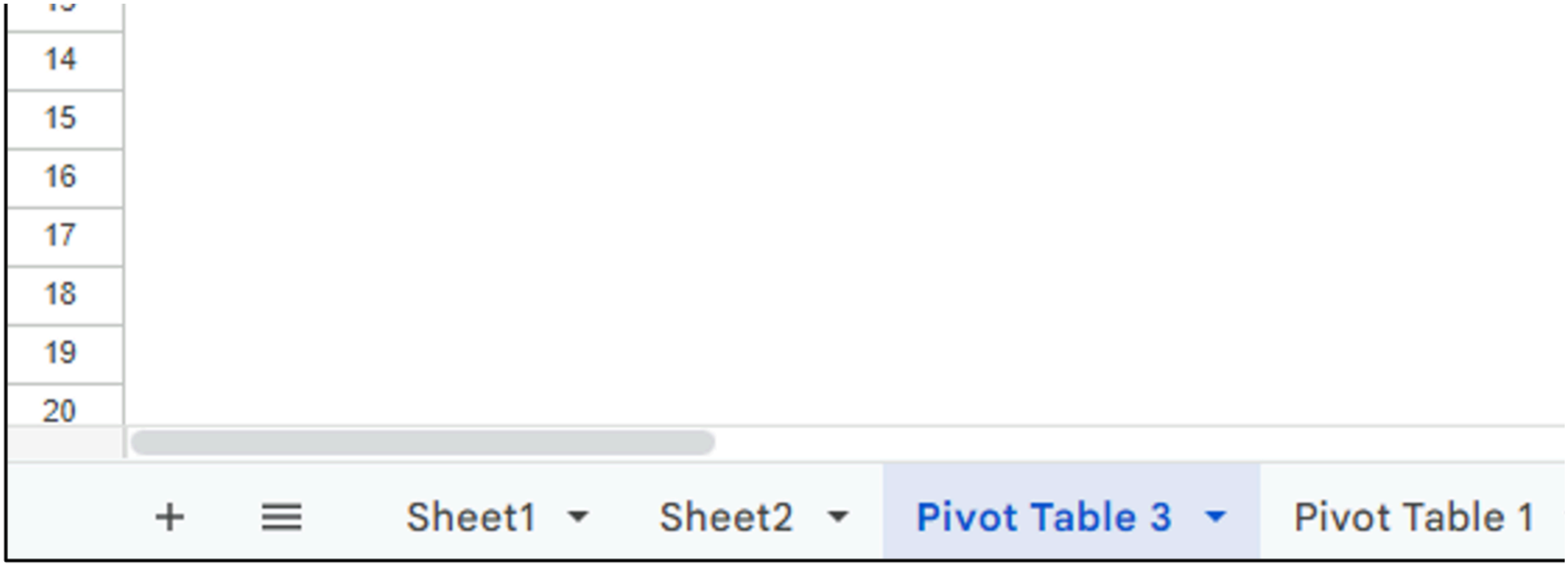
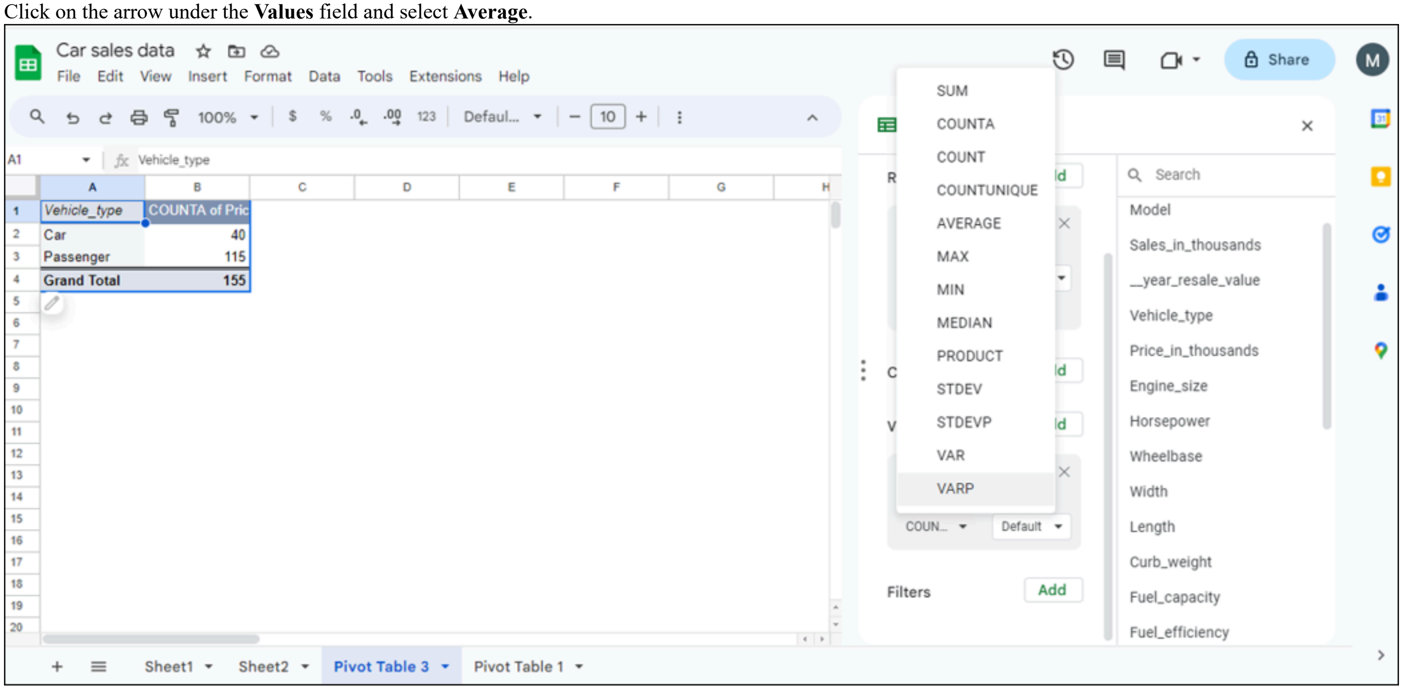
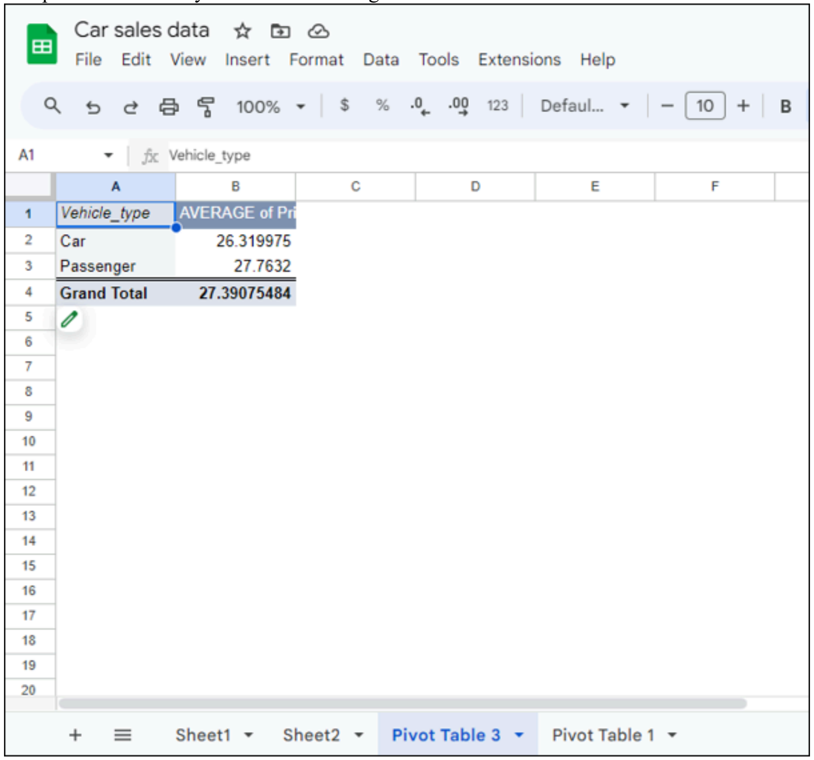


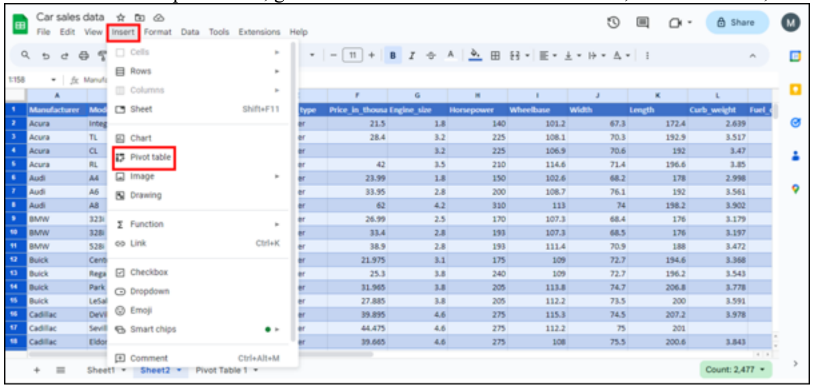
8. Click on the arrow under the **Values** field and select **Average**.



9. The pivot table is ready. You can also change the name of this table.



10. To create one more pivot table, go back to the data. Select the entire data, click on **Insert**, and then click on **Pivot table**.



11. Click on **Create** to start creating the pivot table.